CAPSTONE PROJECT / DELIVERABLE AUDIT

ZentroMart VoiceOps Agent

Implementation report and submission readiness | Version 1.0

STATUS: CODE PACKAGE PREPARED; LIVE AI SCORES, VOICE OUTPUT, PUBLIC REPOSITORY, AND DEMO VIDEO ARE STILL PENDING.

The attached VoiceOps brief calls for four steps: an LLM API baseline, policy-grounded RAG, tool-using conversation handling, and Gnani speech. It also requires a public code repository, a folder-based audio run command, a report of at most four pages, and a short demo. It does not ask the team to train a language model from scratch.

Requirement coverage

Required deliverable	Current implementation	Verification and remaining gap
Step 1 - prompting-only LLM	Configurable OpenAI Responses or Gemini generateContent adapter; separate no-document baseline in evaluate_ai.py.	Provider contract tests pass. No live provider key was present, so the 52-case baseline score is not measured.
Step 2 - RAG	text-embedding-3-small embeddings; cosine top-k over current KB sections plus a matched product row when identifiable; structured GPT answer and evidence IDs.	Ranking, schema, precedence, and catalog citations are tested with mocks. Live embeddings, generation, and RAG score are not measured.
Step 3 - conversations/tools	Existing identity-gated order lookup, one retry, refund math, and mock ticket handover remain deterministic.	Practice set: 96/96 turns pass decisions and actual tool-argument checks.
Step 4 - Gnani voice	Browser voice path plus batch STT file API and Kaveri TTS timbre-v2.5.	Mocked STT/TTS and format tests pass. No fresh provider call or generated audio was possible in this runtime.
Submission run command	python run.py ; writes transcript/reply text, reply audio, and source JSON.	Batch behavior is tested with mocked provider calls. Live audio output still needs a Gnani key.
Report, public code, demo	This report, .env.example, and source package are prepared locally.	No public Git URL or <=2-minute demo video exists yet; add them before submitting.

Assignment source: attached VoiceOps 2026 Capstone Brief, pages 8-14; submission card screenshot supplied by the student.

01 / SYSTEM DESIGN

A small, safe LLM + RAG workflow

Text and voice share the same support agent. Existing deterministic routes handle identity-sensitive order and ticket actions. For the Step 2 comparison, the selected API path embeds and retrieves current policy sections and adds a matched catalog row when a product is identified, then asks the selected LLM for a schema-constrained answer. The app's product answers remain deterministic. If provider access fails, the existing offline BM25 response remains separately labeled.

CUSTOMER TEXT OR AUDIO	DETERMINISTIC ROUTER	RETRIEVE + SELECTED LLM	CITED TEXT / GNANI SPEECH
Voice uses Gnani STT	Verify, calculate, clarify, or use mock tools	Embeddings -> current evidence -> structured response	Gnani Kaveri TTS for voice

Safety and evidence boundaries

- Customer policy answers receive sanitized text, policy excerpts, and recent policy-only context. The Step 2 evaluator may also supply a matched product catalog row. Order records and verification values are never passed as tool data.
- Order lookup, retry limits, calculations, and ticket creation remain deterministic; the model cannot call tools or claim a ticket was created.
- Responses use a strict JSON schema. Returned source IDs are checked against retrieved IDs; unknown citations become an abstention.
- The knowledge-base loader keeps current documents and carries section IDs, document type, status, and effective date through retrieval/output.
- Gnani is the only speech provider. Browser recording is capped at 30 seconds; batch STT accepts WAV, MP3, FLAC, OGG, or M4A up to 60 seconds and 10 MB.

What the assignment genuinely requires

Required	Not required by the final brief
Call an LLM API for a prompting-only baseline; add RAG and compare the same 52 questions; preserve evidence; handle 96 conversation turns; use Gnani STT/TTS; provide a one-command audio batch path and report/demo/public repo.	Training a transformer from scratch, LoRA fine-tuning, a real customer database, live email delivery, or a third-party speech service. The brief permits an LLM API and explicitly reserves speech to Gnani.
Use synthetic data only. Do not package a dataset copy, API keys, .env, or a virtual environment.	The email adapter in this prototype writes only to a local mock outbox; it is optional work, not an external integration.

Provider note: OpenAI and Gemini are supported through LLM_PROVIDER. The current default remains gpt-4.1 for OpenAI; Gemini uses gemini-2.5-flash and gemini-embedding-001 when selected. A live provider key must still be tested.

02 / VERIFICATION

What passed in this workspace

The checks below were run after the code changes. The AI provider paths were mocked because neither API key was present in this task runtime.

Check	Actual result
Dataset validator	PASS - 9 CSVs, 21 knowledge-base documents, order math, joins, evidence, and verification constraints.
Focused unit suite	66 tests passed, including security regressions, Gnani configuration, LLM schema/catalog citation guards, fair Step 1/Step 2 scoring, resumable checkpoints, and the audio-folder writer.
Practice questions	52/52 decisions; 52/52 gold-evidence overlap; 58/58 required facts; 16/16 forbidden facts.
Practice conversations	96/96 decisions; 96/96 actual mock-tool calls and arguments; 82/82 required facts; 32/32 forbidden facts.
Supplied source integrity	37/37 supplied non-cache files match the provided folder by SHA-256; its two existing .pyc cache files also match.
Local HTTP health	PASS in offline mode: 82 policy sections loaded; llm_enabled=false; voice_enabled=false.

The comparison uses identical question correctness for both steps: expected decision plus all required/forbidden rubric facts. RAG evidence hit, exact gold-source coverage, and mean source recall are reported separately; evidence is not counted as a Step 1 failure.

Audit findings and fixes

Before any live score was published, the audit found that the first evaluator version counted missing Step 1 evidence as a failure and checked only the first forbidden-phrase alternative. Both are fixed. The Step 2 benchmark now includes a matched product row when identifiable, and interrupted runs can resume without repeating completed API calls. Package testing also caught a test path that assumed data lived under src and a duplicated tool package; tests now resolve the project root, and the archive uses the supplied tools.

Additional work includes deterministic identity gates, labeled offline fallback, citation validation, and a one-command Gnani audio batch path. These features passed contract tests; live provider behavior remains unverified.

Required API comparison - not yet measured

Step 1 prompting-only	Step 2 RAG	Why not reported
NOT RUN	NOT RUN	No live OpenAI or Gemini key was available in this runtime. The 52/52 deterministic evaluation above is not an LLM baseline and cannot substitute for the required comparison.

Fresh Gnani verification - not yet measured

No transcription or TTS request was made in this final check because GNANI_API_KEY was absent in this runtime. Earlier user-reported browser use is not treated as a fresh verified API result. No audio sample or demo video was fabricated.

The working directory is a local demo only. The mock order tool is not real authentication; sessions and semantic vectors are in memory; email and ticket outputs are local mock files; no public server or real order, email, or ticket service is connected.

03 / BEFORE YOU SUBMIT

Four short tasks remain

The code is packaged locally, but the missing API measurements and public deliverables are required by the brief. Keep keys in the terminal environment; never paste them into chat or place them in the ZIP.

	Task	Action
1	Run the real LLM comparison	In the same PowerShell window where you set the selected provider key, run: `python .\evaluate_ai.py --mode both --output .\ai_eval_results.json` If interrupted, repeat with `--resume`. Use only a JSON result with status complete; copy the real Step 1/2 scores and separate RAG evidence measures into this report.
2	Run Gnani batch and listen	Set GNANI_API_KEY in that session. Put your own short recording (or allowed synthetic test audio) into a folder and run: `python .\run.py .\sample_audio .\sample_outputs` Check the .txt transcript/reply, .wav speech, and .json source IDs.
3	Record the demo	Make a video of at most two minutes: a sourced policy answer with a follow-up, then a human handover or honest abstention. Include the spoken path if it works. Use your own voice; do not record other people without consent.
4	Publish and submit	Upload source code to a public Git repository, include .env.example but no secrets or dataset copy, and add public repository/demo links to the form. The form requires public access to every external link.

Files prepared

The package includes a submission README, a no-secret environment template, application source, tests, and this report. The submission archive intentionally excludes the supplied dataset. It is marked as a draft because the API comparison, live voice sample, public Git URL, and demo video are still outstanding.

Sources: VoiceOps 2026 Capstone Brief, pages 8-14 (provided PDF); OpenAI GPT-4.1 model/API documentation; official Gnani Vachana SDK documentation.